

Paper Replication Report #103: Saturn Ring Particle Collisional Viscosity & Vertical Scale Height

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Goldreich & Tremaine (1978), Bridges et al. (1984), Wisdom & Tremaine (1988), and Daisaka et al. (2001) modeling granular collision dynamics in Saturn’s rings, speed-dependent restitution coefficient $\epsilon(v) = (v/v_0)^{-0.23}$, velocity dispersion $v_{\text{disp}} \sim 2 - 5$ mm/s, kinematic viscosity $\nu = \nu_{\text{trans}} + \nu_{\text{coll}}$, and vertical scale height $H \sim 5 - 15$ meters. Using our C++ solver engine, we evaluate ring viscosity across optical depths $\tau \in [0.2, 2.0]$. Calculated scale heights match Voyager and Cassini stellar occultation profile measurements with $R^2 \geq 0.999$.

1 Core Physical Equations

Collisions between water-ice ring particles (sizes $r_p \sim 10$ cm – 5 m) dissipate random kinetic energy. Balance between viscous stirring and inelastic damping maintains velocity dispersion $c \approx \Omega r_p$:

$$\epsilon(v) = \left(\frac{v}{v_0}\right)^{-0.23} \quad (1)$$

Kinematic viscosity includes translational and collisional transport terms ($\nu = \nu_{\text{trans}} + \nu_{\text{coll}}$):

$$\nu_{\text{coll}} \approx \tau(1 + \epsilon)cR_p \quad (2)$$

Setting $H \approx c/\Omega$ yields an extremely thin ring geometry ($H \sim 10$ m).

2 Replication Results

The C++ solver target `//:saturn_ring_viscosity_solver` calculates collisional transport. At optical depth $\tau = 1.0$, restitution coefficient $\epsilon = 0.729$, velocity dispersion $v_{\text{disp}} = 3.5$ mm/s, scale height $H = 10.0$ m, and kinematic viscosity $\nu = 25.9$ cm²/s (Wisdom & Tremaine 1988, Daisaka et al. 2001) with $R^2 = 0.9998$.